

Duc Nguyen

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SUMMARY

Seasoned engineer with 9+ years building large-scale distributed systems, AI/ML infrastructure, and real-time data platforms on GCP and AWS. Specializes in SaaS companies — defining platform architecture adopted company-wide, driving \$3M+ in infrastructure savings, and delivering systems at 150M–200M+ user scale across telecom, finance, and retail. Proven force multiplier: sets technical direction, aligns cross-functional teams, and accelerates delivery through AI-assisted development including GitHub Copilot and Claude Code CLI.

LANGUAGES & TECHNOLOGIES

Languages: Java (Spring Boot), Python, JavaScript (React/Angular), C++, C# (.NET)

Cloud & Infra: Google Cloud Platform, AWS, Microservices, Distributed Systems, ETL / Streaming Pipelines

AI / ML: LLMs, GenAI, NLP, ASR/STT, Real-Time Transcription, Audio Processing, VoIP · **AI Coding Tools:** GitHub Copilot, Claude Code CLI

PROFESSIONAL EXPERIENCE

Senior Software Engineer — AI Platform | Five9

Remote · June 2021 – Present

- Defined the technical architecture and service contracts for Five9's core AI platform — establishing system boundaries and scaling models adopted across all AI product lines, helping grow the platform from \$100K to \$100M+ in revenue.
- Co-founded Five9's Agentic Coding Guild — defined AI-assisted development standards and tooling adopted org-wide, directly accelerating engineering velocity and reducing ramp-up time for engineers across the AI platform.
- Built and technically led Five9's Transcription team from inception — designed the engineering operating model, architecture standards, and technical roadmap; established the foundational pattern now adopted across 4 AI platform teams.
- Architected the integration framework for Agent Assist's transcription system — standardized interfaces across ASR engines with a low-latency streaming API, enabling Five9 Partner and third-party integrations and unlocking new enterprise customer configurations.
- Drove \$3M+ in annual infrastructure savings through a V1-to-V2 transcription system migration — replacing Google CCAI with Google STT via segmented rollouts to ensure product stability throughout.
- Re-architected Five9's NLP pipeline using an event-driven architecture with distributed caching and optimized async task scheduling — eliminating systemic bottlenecks across the processing chain, reducing end-to-end latency from multi-second to ~100ms, and sustaining throughput at 10k+ concurrent calls in production.
- Architected AQM Prompt Calibration — a multi-tenant LLM evaluation system enabling customer-defined prompt tuning across model providers, with isolated scoring pipelines per tenant; leading a team of 4 engineers through active enterprise onboarding.

Lead Consultant | Virtusa

Irving, TX · March 2020 – June 2021

- Designed the shared microservice decomposition strategy for Citi's Global Risk & Controls platform (200M+ users), establishing cross-cutting service patterns adopted as the migration blueprint by 3+ dependent teams.

Senior API Engineer | Softworld

Plano, TX · July 2019 – March 2020

- Built USAA's Credit Risk Assessment platform integrating proprietary origination systems for 14M+ customers; delivered the Manual Task system resolving complex credit application exceptions end-to-end.

Full-Stack Engineer | Cognizant

Irving, TX · December 2018 – July 2019

- Engineered Macy's ETL framework migrating hundreds of terabytes of SQL catalog data into Cassandra — reducing average query time from 30+ seconds to 1–2 seconds and improving availability across downstream retail systems.

Software Developer | Amdocs

Richardson, TX · March 2017 – December 2018

- Owned DevOps infrastructure for a microservices platform unifying account systems for 150M+ customers; modernized CI/CD pipeline, compressing release cycles from 4 months to 2 weeks.

EDUCATION

B.Sc., Physics & Mathematics — University of Texas at Dallas

GPA: 3.5

Specialized in applied mathematics — linear systems, optimization, and linear algebra; directly informs systems architecture and ASR/NLP pipeline design.